

# Mammalia Raccoon Proximity Network Analysis

<https://github.com/aiyshwary/Raccoon-Network-Analysis>

Python

NetworkX

Jupyter

Graph Analysis

Epidemiology

## OVERVIEW

A graph-based epidemiological study modelling rabies transmission risk in urban raccoon populations through proximity network analysis and centrality measures, supporting public health and wildlife management decisions.

## IMPACT

Constructed and analyzed a raccoon proximity network using real-world contact data, applying centrality measures to identify high-risk individuals and quantify potential rabies spread pathways within urban populations.

## TECHNICAL WALKTHROUGH

Mammalia Raccoon Proximity Network Analysis – Project

### Explanation

#### Project Overview

This project analyzes social interaction patterns among urban raccoons using network analysis to understand how rabies and other pathogens can spread within raccoon populations. Since raccoons are a major vector of rabies, studying how often and how closely they interact helps estimate disease transmission risk and supports better public health and wildlife management strategies.

### Objective

#### Model raccoon interactions as a graph/network

Identify high-risk raccoons that can spread pathogens quickly Use centrality measures to understand social structure and disease flow

#### Dataset & Network Construction

24 raccoons → represented as nodes

~2000 interactions → represented as weighted edges

#### Data collected using proximity-detecting collars

Edge weight = time spent in close proximity Monthly interaction data used to build the network

#### If two raccoons are connected, it means they had close physical

interaction, which is critical for rabies transmission.

Why Network Analysis?

Disease spread is not random.

### It depends on

i) Who interacts with whom

ii) How frequently

iii) Who acts as a bridge between groups

iv) Network analysis helps identify super-spreaders and critical connectors.

## Centrality Measures Used & Interpretation

### Degree Centrality

i) Measures number of direct connections

ii) High degree = more interactions

### Inference

i) Raccoon #4 has the highest degree

ii) Acts as a high-contact individual

iii) High risk of initiating widespread infection

### Closeness Centrality

Measures how close a node is to all others

**Indicates how fast something can spread**

### Inference

i) Raccoon #2 has highest closeness centrality

ii) Can spread pathogens rapidly across the entire network

### Betweenness Centrality

Measures how often a node lies on shortest paths

**Identifies bridge nodes**

### Inference

i) Raccoon #16 connects different subgroups

ii) Acts as a key transmission bridge

iii) Removing or vaccinating this raccoon can significantly slow disease

iv) spread

v) Clustering Coefficient

vi) Measures how tightly a raccoon is embedded in a local group

vii) Indicates family or group behavior

### Inference

i) Raccoon #2 has highest clustering

ii) Infection here may cause localized outbreaks within families

### Eigenvector Centrality

Measures influence based on connections to influential nodes

### Inference

i) Raccoon #4 again ranks highest

ii) Connected to other highly interactive raccoons

iii) Considered a super-spreader candidate

iv) PageRank Algorithm

v) Evaluates overall importance considering direction and weight of

vi) interactions

vii) Helps rank raccoons by global influence

**viii)** Key Findings

**ix)** Raccoon populations are highly interconnected

**x)** Disease can spread much faster than expected

**xi)** A few raccoons play disproportionately large roles in pathogen

**xii)** transmission

**xiii)** Targeted intervention (vaccination/tracking of key raccoons) is more

**xiv)** effective than random control

**xv)** Technologies & Concepts Used

**xvi)** Graph Theory

**xvii)** NetworkX (Python)

**xviii)** Weighted graphs

## Centrality algorithms

## Epidemiological modeling concepts

### KEY DETAILS

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**1.** Built a weighted proximity network from the mammalia-raccoon-proximity dataset using Python and NetworkX.

**2.** Computed degree, betweenness, closeness, and eigenvector centrality to rank individuals by transmission risk.

**3.** Identified superspreader nodes whose removal would most disrupt disease propagation chains.

4. Visualized network topology and centrality distributions to communicate findings clearly.

5. Results provide evidence-based recommendations for targeted wildlife intervention strategies.